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Fish Growth Model for Nile Tilapia (*Oreochromis niloticus*) in Wastewater Oxidation Pond, Thailand

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Abstract

The bioenergetics fish growth model of Nile Tilapia (*Oreochromis niloticus*) cultured in the cages in wastewater oxidation pond is presented. The oxidation pond is located in the very successful project area “The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province” in Thailand. There are five oxidation ponds connected in series in the research area. The measurement of Nile tilapia growth is taken in the third pond in which the water quality is in the effluent standard. The water quality and fish weight are measured in each month for one year period. The fish growth model in the form of ordinary differential equation is introduced to understand the behavior of fish growth due to various environmental factors which are dissolved oxygen demand (DO), water temperature, concentration of plankton, ammonia and biochemical oxygen demand (BOD). In this presented model, we have introduced a new form of food assimilation efficiency as a function of fish weight. It is found that the predicted fish weight obtained from the model is in good agreement with the measurements. Also, this presented model can be applied to predict the fish weight in a wastewater stabilization pond when environmental factors have been changed.

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Keywords: Growth; Model; Nile tilapia; Fish; Wastewater; Oxidation; Pond; BOD; DO

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1. Introduction

The King's initiative on community wastewater treatment under the nature-supporting-nature processes has been shown the success through oxidation pond techniques as aquatic ecosystems. Practically, wastewater from the city of Petchaburi was drained through 0.45 m pipe at approximately 18.5 km to reach the experimental site. Out-pipe wastewater transferred to sedimentation pond with 5 days retention time, then run over to pond number 2, 3 and 5 with retention time of 13, 7 and 7 days, respectively. The treatment efficiency of the whole systems was found at 85 percent of BOD and COD decreasing, but more efficiency for coliform bacteria reducing as well as TDS, SS, heavy metals and plant nutrients. Experimental results also indicated high growth of phytoplankton and zooplankton in which they were good enough for herbivore fish aquaculture by allowing 3 fishes per sqm of oxidation pond surface. Luckily, the fish product could make benefits more than a half million Bath per year, and then dividing to support 50 percent for community, 30 percent for two primary schools, 10 percent for elderly people, and the last 10 percent for project staff security. The fish products have been noticed since 1992, and found that the product gained were not consistency. After close study, it came out with requiring to investigate the relationship between fish growth and concentration of plankton as the same as DO, BOD, NH_3 , and water temperature in the oxidation ponds.

Actually, fish growth model for Nile tilapia (*Oreochromis niloticus*) was used to study in case of clean freshwater, not in wastewater oxidation pond. It is expected that fish growth model in freshwater model would be applicable more to wastewater, especially in wastewater treatment systems likewise oxidation pond as belonged to the Royal project in Petchaburi, Thailand. Summarily speaking, the objectives of this study are aimed to find out the feasible application of fish growth model in wastewater, and also the involved factors for Nile tilapia growth.

2. Methodology

An experiment was conducted in the community-based wastewater from Municipality Phetchaburi located in the very successful project area "The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province" in Thailand (Fig. 1.). The community wastewater treatment system have sequence 5 ponds, such as a sedimentation pond, three oxidation ponds and a polishing pond, surface area of 3, 7.5, 11.4, 7.1 and 8.9 ha., respectively. Practically, wastewater from the city of Phetchaburi was drained through 0.45 m pipe at approximately 18.5 km to reach the experimental site.

The Nile tilapia (*Oreochromis niloticus*) cultured in the net cages fixed in the oxidation ponds no.2 (surface area of 11.4 ha.) because it was found the highest growth and optimal for water quality. The net cages filled at the density of 1, 2, 3, 4 and 5 fishes per sqm, each density have three replications and cultured for 48 weeks without feeding. The net cages were surface area of 25 m² (5x5 m²) and submerge 1.5 m. The stocking size of Nile tilapia cultured in the net cages for average total length of 7.0 ± 0.33 , 7.1 ± 0.3 , 57.1 ± 0.31 , 6.8 ± 0.39 and 6.8 ± 0.44 cm, respectively, and average body weight of 5.99 ± 0.75 , 5.99 ± 0.8 , 25.90 ± 0.75 , 5.34 ± 0.93 and 53.4 ± 10.1 g, respectively. The cultured fish was sampling for measuring and weighting of their total length and body weight, respectively at every 4 weeks, and analysed data as the specific growth rate, correlation coefficient values and coefficient of condition were calculated.

During the experiment, the following parameters of water quality in cages, i.e., temperature, pH, dissolved oxygen demand (DO), biological oxygen demand (BOD), ammonia (NH_3), nitrate (NO_3^-), alkalinity, hardness, total-phosphate, suspended solid (SS) and concentration of plankton (natural food)

were determined and recorded. Both sampling and analyzed water quality were procedure in the American Public Health Association (APHA) [1].



Fig. 1. Map of The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province in Thailand

3. Modelling and characterization

3.1. Modelling development

The growth model of fish was developed by Ursin [2] and modified by Bolte et al. [3], it was a bioenergetics growth model representing the anabolism and catabolism of fish processes. The rate of change of fish weight can be expressed by

$$\frac{dW}{dt} = HW^m - kW^n \quad (1)$$

where, W is fish weight (g), t is time (day), H is the coefficient of net anabolism ($\text{g}^{1-m} \text{day}^{-1}$), m is the exponent of body weight for net anabolism, k is the coefficient of fasting catabolism ($\text{g}^{1-n} \text{day}^{-1}$), and n is the exponent of body weight for fasting catabolism.

The growth rate of fish increased by anabolism process comprised of feed rate or food assimilation in term of daily ration, but its growth rate is decreased by catabolism process in term of feeding and fasting catabolism, Ursin [2]. Then equation (1) can be written in the form of daily ration as

$$\frac{dW}{dt} = \left(b \frac{dR}{dt} - ab \frac{dR}{dt} \right) - kW^n \quad (2)$$

$$\frac{dW}{dt} = b \frac{dR}{dt} - \left(ab \frac{dR}{dt} + kW^n \right) \quad (3)$$

where, $\overline{dt}$ is the daily ration (g day^{-1}), a is the fraction of the food assimilated that is used for feeding catabolism (dimensionless), and b is the efficiency of food assimilation (dimensionless).

The right hand side of equation (3) represents the gross anabolism, feeding catabolism and fasting catabolism, respectively. Fish weight increases by anabolism process that is directly proportional to the daily ration. In contrast, weight of fish decreases by feeding catabolism and living activities.

Rewriting equation (3), we obtain

$$\frac{dW}{dt} = (1 - a)b \frac{dR}{dt} - kW^m \quad (4)$$

The parameter b means food assimilation efficiency that refers to the proportion of the gross energy or food intake available for metabolizable energy, Nath et al. [4], and that is typically not constant. In this work, we propose a new function of food assimilation efficiency as a function of fish weight as:

$$b = d - [c / (1 + (e^{-\varepsilon(W - Tc)}))] \quad (5)$$

The range of b values is $d - c \leq b \leq d$ which refers to the minimum and maximum food assimilation efficiency. It is not a constant along time simulated. It is totally different from the parameter value as described in Yi [5]. It should be noted that ε is a parameter meaning the rate of change from its maximum to minimum value while Tc is a time period to shift the rate of change of b .

The feeding rate of fish was depended on body weight and environmental factors, see Brett [6], such as temperature (T) in Brett et al. [7], dissolved oxygen demand (DO) in Stawart et al. [8], Unionized ammonia (A) in Colt and Tchobanoglous [9], availability of feed in Brett [10] and biochemical oxygen demand (BOD). So, these factors affect directly to the growth rate of fish and feeding, see Brett [6] and Cuenco et al. [11]. Hence, the daily ration would depend on the environmental factors, we assume it in the form of

$$\frac{dR}{dt} = \tau \delta \nu \beta f h W^m \quad (6)$$

where, τ is the temperature factor ($0 < \tau < 1$, dimensionless), δ is the DO factor ($0 < \delta < 1$, dimensionless), ν is the unionized ammonia factor ($0 < \nu < 1$, dimensionless), f is the relative feeding level ($0 < f < 1$, dimensionless), β is the BOD factor ($0 < \beta < 1$, dimensionless) and h is the coefficient of food consumption ($\text{g}^{1-m} \text{day}^{-1}$). The daily ration function has included the BOD factor in the model to predict fish growth in the stabilization pond of our research area.

Brett [7] presented that the food consumption of fish species increases when the water temperature (T) increases constantly from the lower limit ($T_{\min}$) to the optimum (T_{opt}). On the other hand, the food consumption decreases when the optimum temperature increases to the maximum temperature ($T_{\max}$). The temperature influenced for food consumption and anabolism was proposed by Svirezhev et al. [12] and Bolte et al. [3]. The temperature effect is written in term of function (τ) as shown in equations (7, 8). The behaviour of this function is shown in Fig. 2. (a).

$$\tau = \exp\{-4.6 [(T_{\text{opt}} - T) / (T_{\text{opt}} - T_{\min})]^4\} \quad \text{if } T < T_{\text{opt}} \quad (7)$$

$$\tau = \exp\{-4.6 [(T - T_{\text{opt}}) / (T_{\max} - T_{\text{opt}})]^4\} \quad \text{if } T \geq T_{\text{opt}} \quad (8)$$

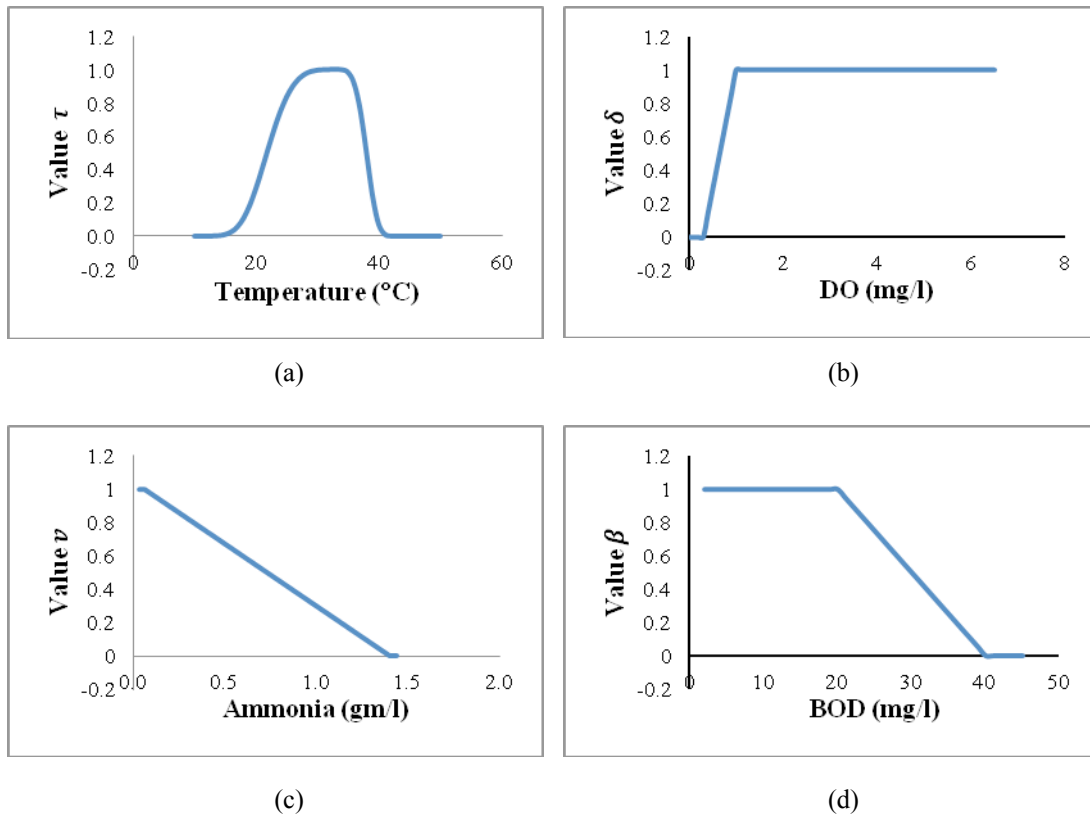


Fig. 2. Patterns of environmental factors that affect to the food consumption of fish: (a) τ is the temperature factor ($0 < \tau < 1$, dimensionless), (b) δ is the DO factor ($0 < \delta < 1$, dimensionless), (c) v is the unionized ammonia factor ($0 < v < 1$, dimensionless) and (d) β is the BOD factor ($0 < \beta < 1$, dimensionless).

Ursin [2] and Sperber et al. [13] assumed that the coefficient of catabolism (k) increases exponentially with temperature and this exponential form modified by Nath et al. [4] to include the minimum temperature (assumed to be equivalent to $T_{\min}$) that fish cannot survive as in equation (9).

$$k = k_{\min} \exp [j (T - T_{\min})], \quad (9)$$

where $k_{\min}$ is a coefficient of fasting catabolism ($\text{g}^{1-n} \text{day}^{-1}$) at $T_{\min}$, and j is a constant to describe temperature on catabolism (1°C^{-1}).

The DO concentration is the dominant factor affecting the food consumption of fish. Bolte et al. [3] described the effects of DO concentration on food consumption and anabolism in term of function δ as shown in equations (10, 11, 12). The behaviour of this function is shown in Fig. 2. (b).

$$\delta = 1.0 \quad \text{if } \text{DO} > \text{DO}_{\text{crit}} \quad (10)$$

$$\delta = (\text{DO} - \text{DO}_{\min}) / (\text{DO}_{\text{crit}} - \text{DO}_{\min}) \quad \text{if } \text{DO}_{\min} \leq \text{DO} \leq \text{DO}_{\text{crit}} \quad (11)$$

$$\delta = 0.0 \quad \text{if } \text{DO} < \text{DO}_{\min} \quad (12)$$

Colt and Armstrong [14] and Cuenco et al. [11] described the unionize-ammonia (A) that was also affected to food consumption. That is, the food consumption was not affected when the unionize-ammonia is less than a critical limit (A_{crit}) but the food consumption decreases when the unionize-ammonia increases. Bolte et al. [3] described the effects of unionize-ammonia on food consumption in term of function v as shown in equation (13, 14, 15). The behaviour of this function is shown in Fig. 2. (c).

$$v = 1.0 \quad \text{if } A < A_{crit} \quad (13)$$

$$v = (A_{max} - A) / (A_{max} - A_{crit}) \quad \text{if } A_{crit} \leq A \leq A_{max} \quad (14)$$

$$v = 0.0 \quad \text{if } A > A_{max} \quad (15)$$

The BOD concentration also affects the food consumption of fish. In fact, BOD is related to DO. If BOD is below the critical limit (BOD_{crit}), fish gets benefit in food consumption but the food consumption decreases when BOD increases. If the value of BOD concentration is greater than the critical value (BOD_{max}), fish cannot survive. We give the function β to represent the BOD effects. It is shown in equations (16, 17, 18). The behaviour of this function is shown in Fig. 2. (d).

$$\beta = 1.0 \quad \text{if } BOD < BOD_{crit} \quad (16)$$

$$\beta = (BOD_{max} - BOD) / (BOD_{max} - BOD_{crit}) \quad \text{if } BOD_{crit} \leq BOD \leq BOD_{max} \quad (17)$$

$$\beta = 0.0 \quad \text{if } BOD > BOD_{max} \quad (18)$$

The natural food in ponds such as phytoplankton and zooplankton is directly proportional to the initial concentration of limiting nutrient, see Parsons and Takahashi [15]. The relative feeding level (f) with potential net primary productivity and standing crop of Nile tilapia is proposed by Ivlev [16]. The feeding level function is shown in equation (19),

$$f = \frac{r}{R} = 1 - \exp \left[-s \left(\frac{P}{B} \right) \right], \quad (19)$$

where, r is the actual daily ration (g day^{-1}), R is the maximal daily ration (g day^{-1}), s is the proportionality coefficient of potential net primary productivity to natural food (dimensionless), P is the potential net primary productivity ($\text{g m}^{-3} \text{ day}^{-1}$) and B is the standing crop of Nile tilapia (g m^{-3}). Following Ivlev [16], we defined P as the concentration of natural food and B as the number of fish in this work.

Finally, the growth rate ($\frac{dW}{dt}$) of Nile tilapia can be expressed by

$$\frac{dW}{dt} = \{(1 - a)bt\delta v\beta f W^m\} - kW^n \quad (20)$$

We solve numerically the fish growth model in equation (20) by the classical Runge-Kutta method by varying various environmental factors to investigate the growth rate. Model parameters can be evaluated by referring to some existing literatures and fitting approach.

3.2. Parameter evaluation

Nath et al. [4] analyzed oxygen consumption data of fasting Nile tilapia and estimated the mean of n to be 0.81. Ursin [2] suggested that the exponent m is approximately 0.67. These two values are used in this presented model.

Nath et al. [4] reported the Nile tilapia feeding in ponds. It was assumed that the efficiency of food assimilation (b) is 0.62 but, in this work, we propose equation (5) to represent b . The value of parameters a and h for Nile tilapia feeding on natural food were assumed to be 0.53, see Nath et al. [4] and 0.8, see Bolte et al. [3], respectively.

The results from experiment in laboratory for Nile Tilapia reveal that $T_{\min}$, T_{opt} and $T_{\max}$ are appear to be 15 °C, see Gannam and Phillip [17], 33 °C, see Caulton [18], and 41 °C, see Denzer [19], respectively. Nath et al. [4] and Yi [5] used data on fasting Nile tilapia from Satoh et al. [20] to estimate the values of $k_{\min}$ and j . It is found that they are 0.00133 and 0.0132 respectively.

Boyd [21] described that Nile tilapia can tolerate in case of low DO concentration and can survive in the environment where other few species can survive, except air breathing species, see Yi [5]. Nile tilapia was able to use atmospheric oxygen when DO concentration drops less than 1 mg l⁻¹, Chervinski [22]. Ahmed and Magid [23], Magid and Babiker [24] described that Nile tilapia can survive in the case of very low DO concentration ranging from 0.1 to 0.3 mg l⁻¹ while the values of DO_{crit} and $\text{DO}_{\min}$ have not been reported. Teichert-Coddington and Green [25] presented that the practical threshold DO for Nile tilapia was not greater than 10 % of saturation. Yi [26] presented that the values of DO_{crit} and $\text{DO}_{\min}$ for caged Nile tilapia are 1.0 and 0.3 mg l⁻¹ respectively. These two values are used in our presented model.

Abdalla [27] investigated the effect of ammonia for Nile tilapia growth and determined that $A_{\max}$ and A_{crit} are 1.4 and 0.6 mg l⁻¹ respectively.

3.3. Data requirement for model validation

Almost all values of parameters used in our simulations are carried out from the literatures shown in Table 1.

Table 1. Parameter values used in the simulations and their data sources.

Parameters	Values used in simulations	Sources
a	0.53	Nath et al. [4]
DO_{crit}	1.0	Teichert-Coddington and Green [25]
$\text{DO}_{\min}$	0.3	Ahmed and Magid [23] Magid and Babiker [24]
h	0.8	Bolt et al. [3]
j	0.0132	Nath et al. [4]
$k_{\min}$	0.00133	Nath et al. [4]
m	0.67	Ursin [2]
n	0.81	Nath et al. [4]
s	21.38	Calibration
$T_{\min}$	15	Gannam and Phillip [17]
$T_{\max}$	41	Denzer [19]
T_{opt}	33	Caulton [18]
A_{crit}	0.06	Abdalla [27]
$A_{\max}$	1.40	Abdalla [27]
BOD_{crit}	20	experiment
$\text{BOD}_{\max}$	40	experiment

4. Results

4.1. Experiment

The experimental results reported in this work are for the density of one fish per sqm. The average total length and body weight of Nile tilapia cultured in the cage at the density of one fish per sqm are 7.0 ± 0.33 cm and 5.99 ± 0.75 g, respectively. We have recorded for 48 weeks without feeding. The results showed that Nile tilapia gradually grew all the time to the average total length of 28.3 ± 1.85 cm and the average total weight of 433.33 ± 86.40 g. The body weight was measured in every 4 weeks as shown in Table 2.

The average daily growth and specific growth rate for Nile tilapia are relatively high showing fatty fish, see [28], [29], [30]. These values are greater than some types of natural fish in Thailand, see Sidthimunka [31].

Table 2. The average body weight (g) of Nile tilapia (*Oreochromis niloticus*) cultured in cages for the density of 1 fish per sqm in oxidation pond at The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province in Thailand

Weeks	Body weight (g)
0	5.57 ± 0.75
4	24.46 ± 5.11
8	65.42 ± 12.44
12	113.61 ± 21.73
16	157.44 ± 27.96
20	195.92 ± 37.95
24	243.00 ± 46.04
28	273.22 ± 49.48
32	312.76 ± 56.60
36	338.12 ± 59.93
40	343.88 ± 65.79
44	400.81 ± 78.93
48	433.33 ± 86.40

Throughout the experiment for the cages of one fish per sqm, the indicators of water qualities are composed of temperature, pH, DO, BOD, ammonia, nitrate, alkalinity, hardness, total phosphate, suspended solid and concentration of plankton (natural food). These values are shown in Table 3. It is shown that the water qualities are appropriate for fish growth, excepting the values of ammonia and nitrate that are above the standard values. However, Nile tilapia could tolerate and assimilate in the environment of high ammonia and nitrate levels and these values are non-toxic to fish. So, the growth of Nile tilapia increases continually from the beginning of culture until the 36th week.

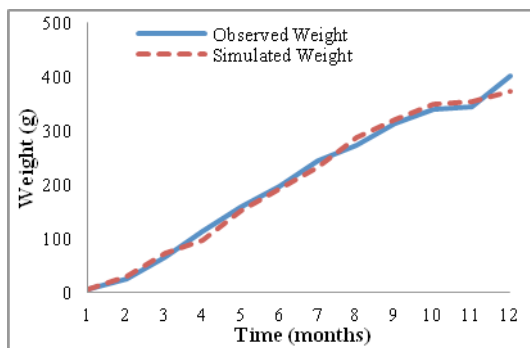
Table 3. The average values of water qualities in cages of 1 fish per sqm in the oxidation pond at The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province in Thailand

Parameters	Unit	experiment
Temperature	°C	29.7 ± 2.06
pH	-	8.3 ± 0.40
DO	mg/l	9.6 ± 3.41
BOD	mg/l	24.50 ± 7.64
SS	mg/l	68.65 ± 21.39
Alkalinity	mg/l	121 ± 19.16
Hardness	mg/l	171 ± 38.77

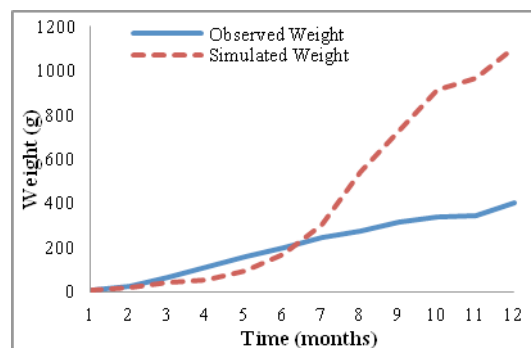
Total-Phosphate	mg/l	1.90 ± 0.66
Nitrate	mg/l	1.10 ± 0.74
Ammonia	mg/l	0.83 ± 0.48
Plankton	Cells/m ³	$1,580,360 \pm 288,779$

4.2. Simulations

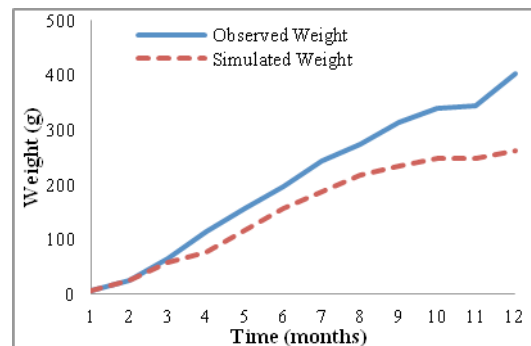
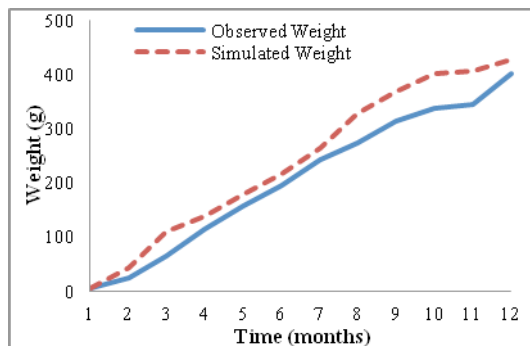
We have simulated the growth curve and compared with some observed data of fish growth cultured in the cages for the density of 1 fish per sqm in the oxidation ponds. We set the food assimilation efficiency as a function of fish weight as shown in equation (5). Here $d = 0.9$, $c = 0.7 = 0.05$, and $t = 130$ days. The prediction of fish weight is shown in Fig. 3. (a). They are in good agreement with the observed data. This shows the accuracy of our presented model and the parameter values employed in the simulation. When we set the value of b as a constant 0.62 which is the same value as reported in (Nath et al.), the simulation results are completely incorrect, see in Fig. 3. (b). This shows that we should set the food assimilation efficiency as a function of fish weight or time instead of a constant. We have also shown the applicability of our model when the BOD effect is included. We set the BOD values on the range 10-20 mg l⁻¹. The water quality in this case is better than those values that we have measured in the experimental site. The simulation of fish weight is shown in (Fig. 3. (c)). It is higher than the observed data meaning that fish has higher rate of growth due to lower BOD. However, if we increase the values of BOD in the range of 25-30 mg l⁻¹, the fish weight from the simulation, see Fig. 3.(d), is less than the observed data. It is reasonable because we have simulated the fish growth in the environment of worse water quality.



(a)



(b)



(c)

(d)

Fig. 3. Comparison of simulated Nile tilapia growth with observed data in the oxidation pond at The Leam Phak Bia Environmental Study Research and Development Project under Royal Initiatives Petchaburi Province in Thailand: (a) b is not constant as a function of fish weight and BOD observed, (b) b is a constant 0.62 and BOD observed, (c) b is a function of fish weight and BOD is of 10-20 mg l^{-1} , (d) b is a function of fish weight and BOD is of 20-30 mg l^{-1} .

5. Conclusions

The community wastewater treatment system has been shown the success through oxidation pond techniques as aquatic ecosystems. The treatment efficiency of the whole systems was found at 85 percent of BOD decreasing, but more efficiency for coliform bacteria reducing as well as TDS, SS and plant nutrients. Experimental results indicate high growth of plankton in which they were good enough for herbivore fish culture in an oxidation pond.

The experiment cultured Nile tilapia at the density of one fish per sqm in the oxidation pond number 2, growth rate of fish was measured for 48 weeks and some important environmental factors was also collected for studying the growth rate that responds to that environmental factors. We have also presented the mathematical model of fish growth based on the bioenergetic model. Here we have included both the BOD and ammonia effects. These two additional factors should be included in the presented model because the fish growth is observed in the oxidation pond. Moreover, we have presented a new form of food assimilation efficiency. It is represented in term of a fish weight function that is not traditionally a constant along the simulation time. By this setting, the values of predicted fish weight are in good agreement with the observed data. Thus, we can apply this presented model to investigate fish growth in oxidation pond when environmental factors change.

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